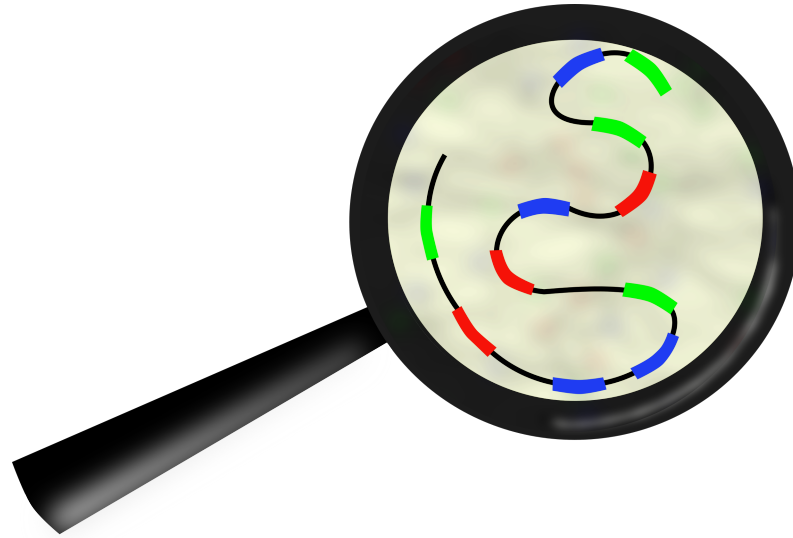
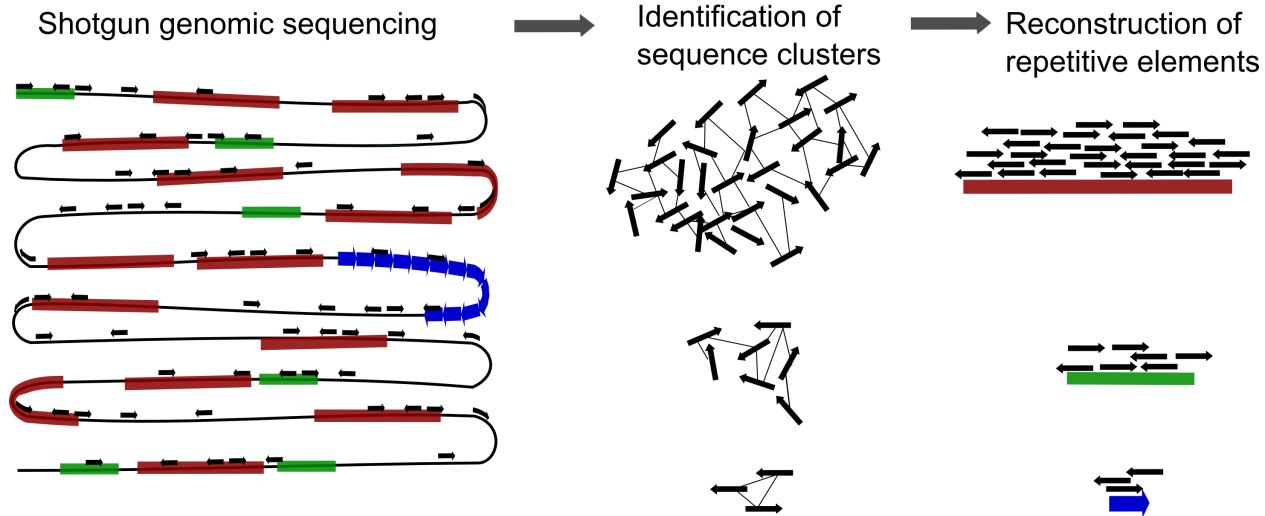


RepeatExplorer tools for genome annotation



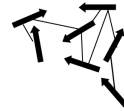
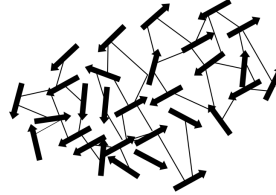
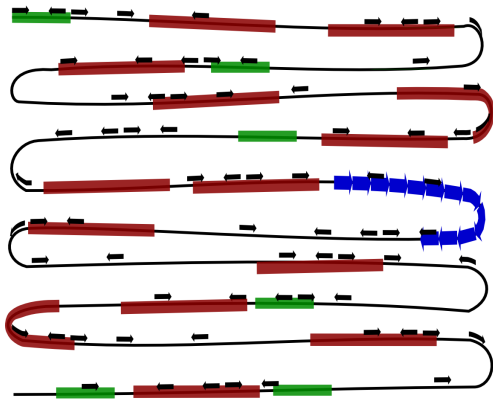
RepeatExplorer tools for genome annotation



Characterization of repeat from low-pass shogun sequencing

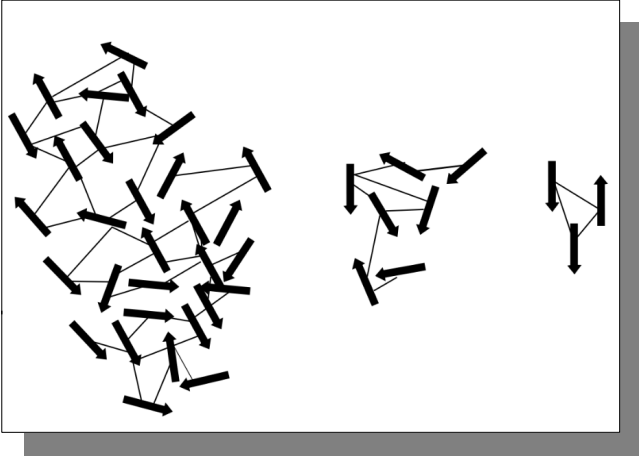
RepeatExplorer tools for genome annotation

Genome Assembly



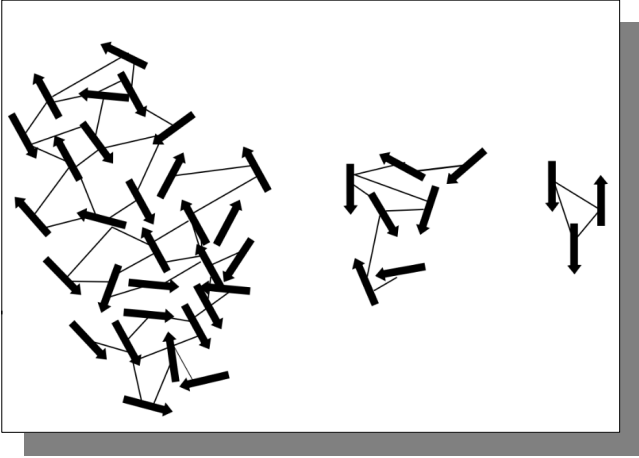
RepeatExplorer server tools for genome annotation

Profrep



RepeatExplorer server tools for genome annotation

Profrep

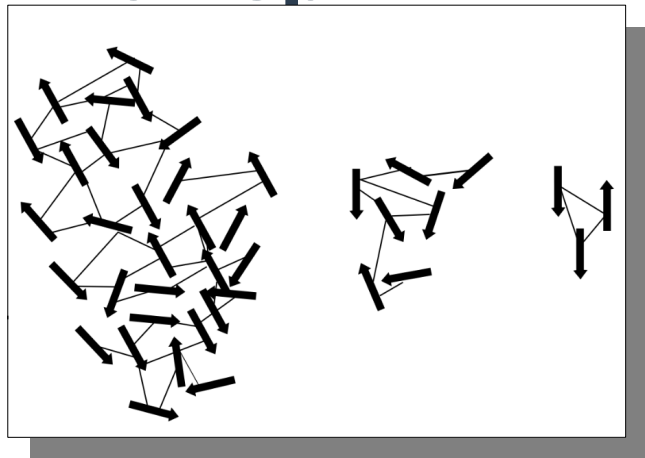


DANTE

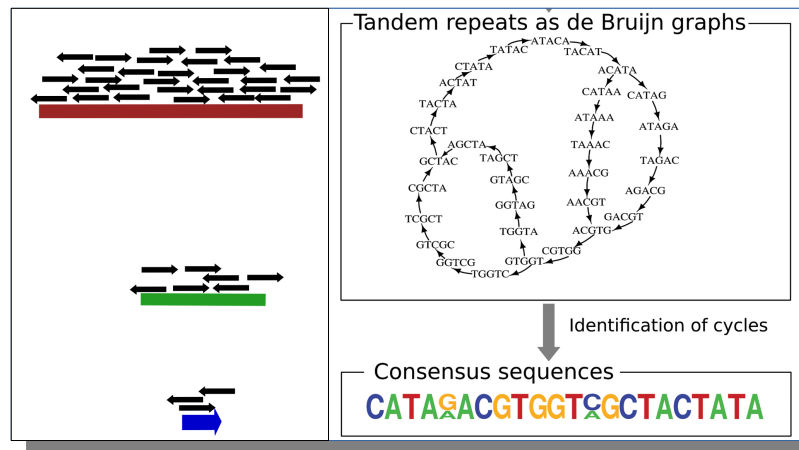


RepeatExplorer server tools for genome annotation

Profrep



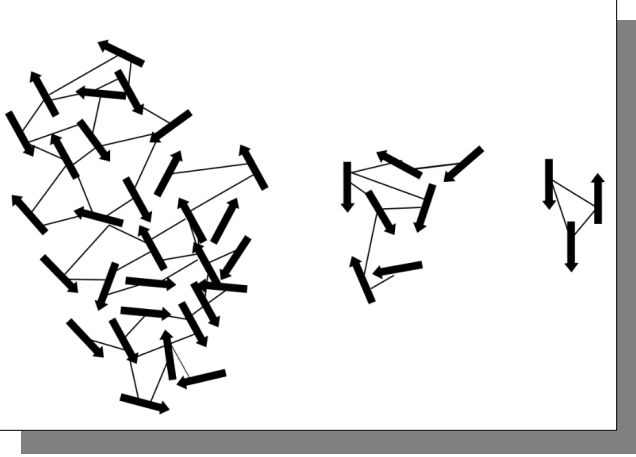
DANTE



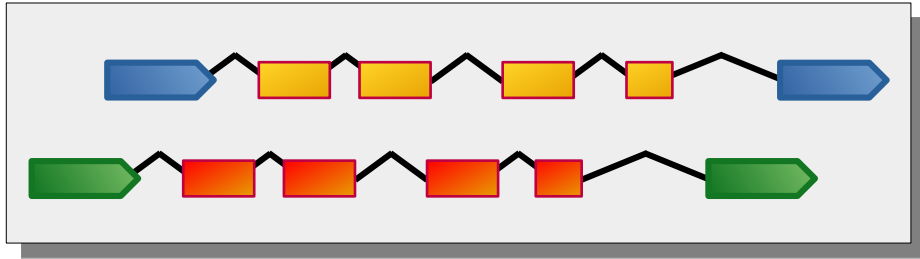
Library based annotation

RepeatExplorer server tools for genome annotation

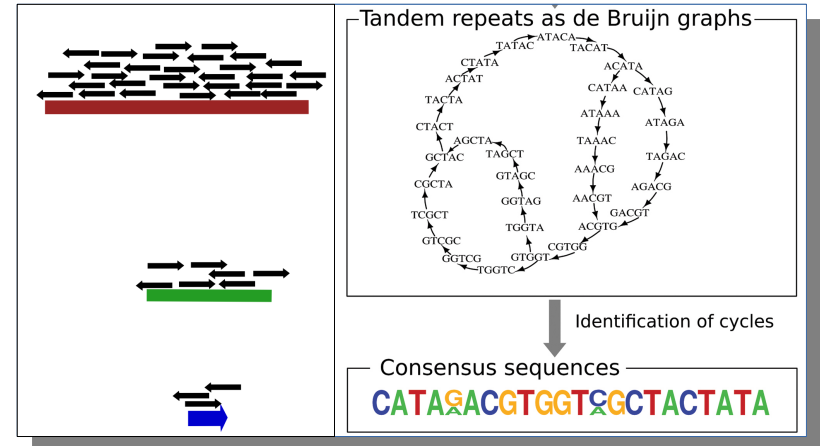
Profrep



DANTE

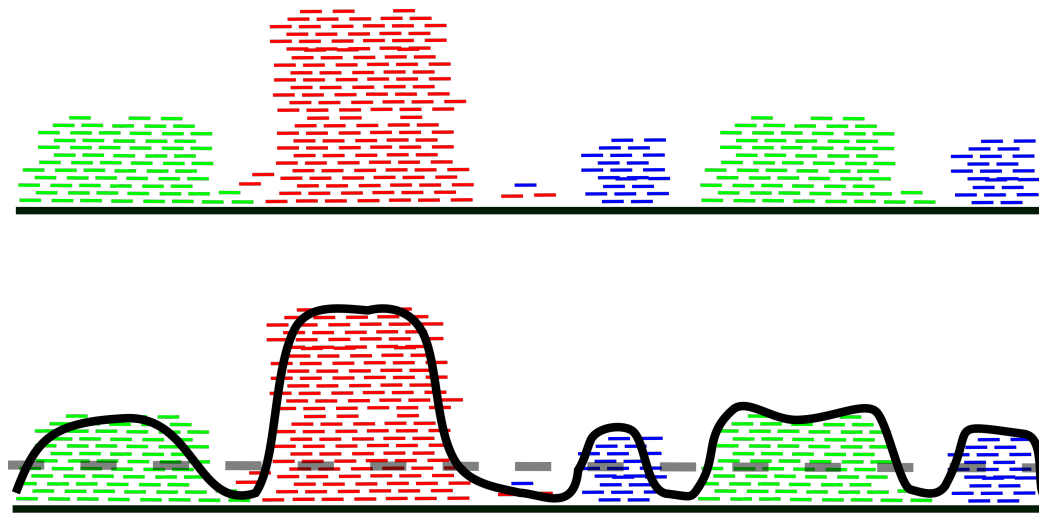
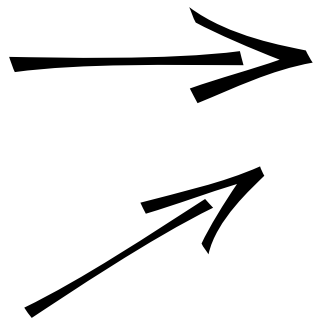
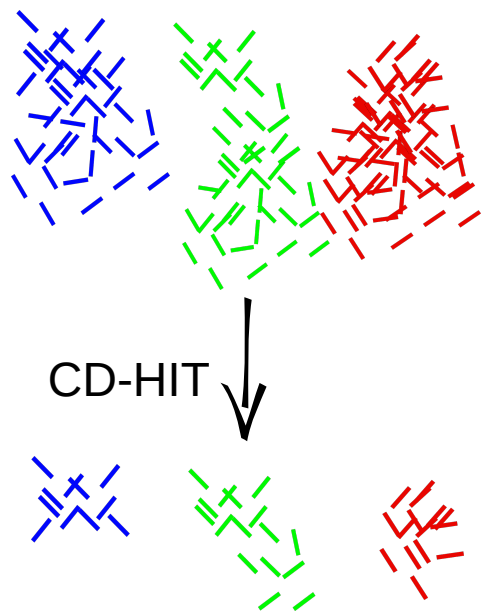


DANTE LTR



Library based annotation

PROFREP

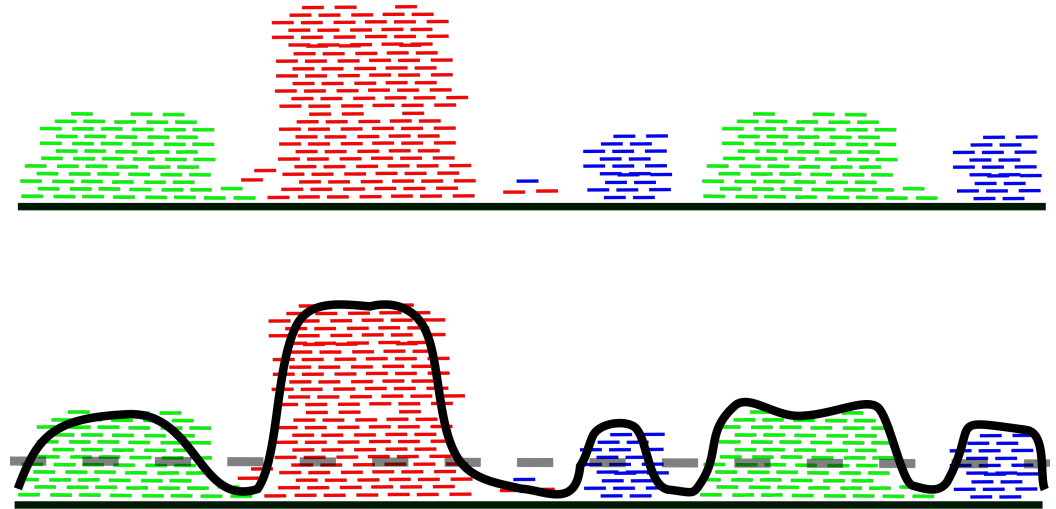
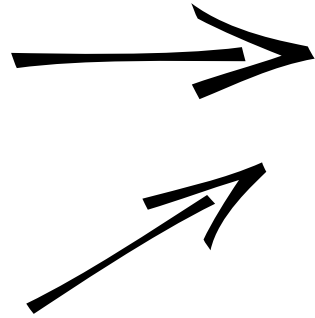
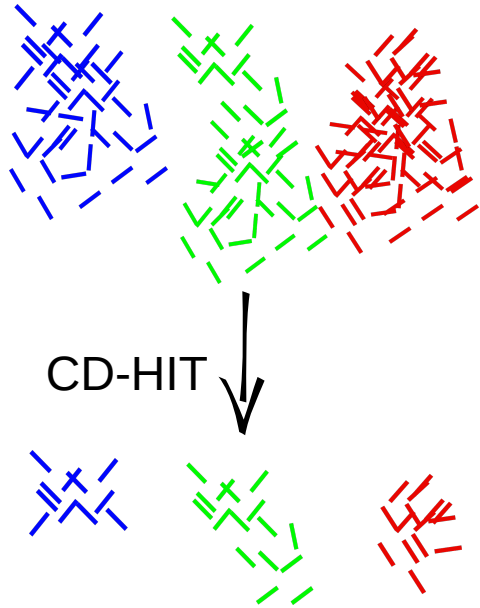


Input

- Assignment to clusters (hitsort.cls)
- Reads
- Cluster annotation – curated
- Genome size (optional)



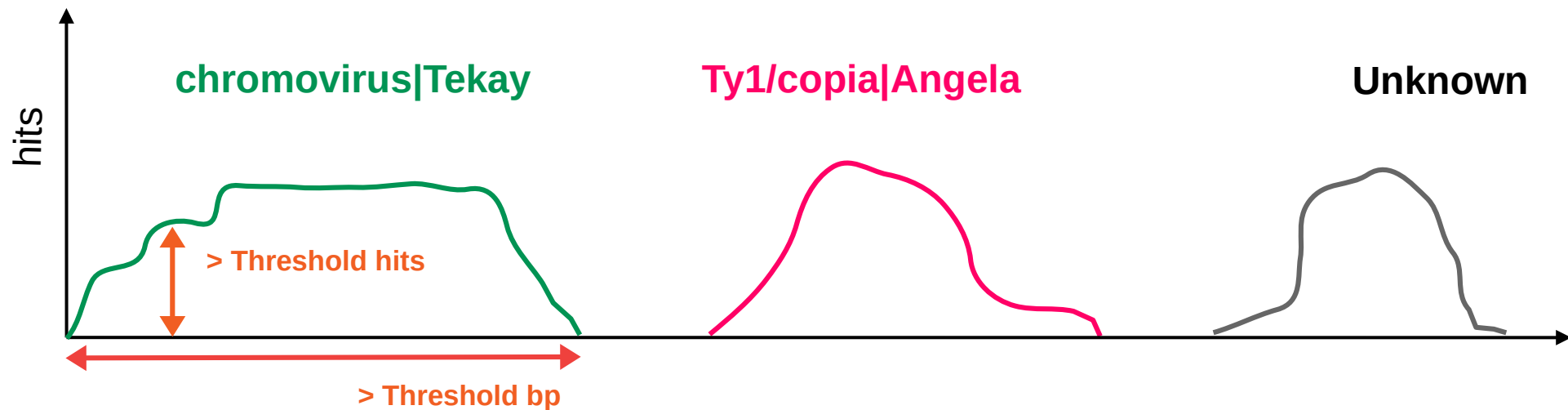
PROFREP



- Slow – One to many mapping
- Copy Number information



PROFREP



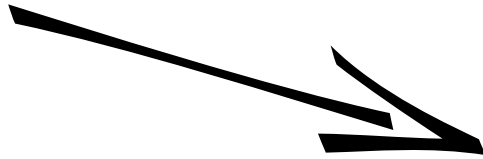
##gff-version 3

```
9 profrep repeat 53 1221 . . . Name=repeat|mobile_element|Class_I|LTR|Ty1/copia|SIRE;Average_PID=83
9 profrep repeat 7819 7921 . . . Name=repeat|mobile_element|Class_I|LTR|Ty3/gypsy|chromovirus|Tekay;Average_PID=77
9 profrep repeat 8365 8449 . . . Name=unknown_CL203;Average_PID=75
9 profrep repeat 9353 9434 . . . Name=unknown_CL209;Average_PID=83
```

Library Based Repeat Annotation



RepeatMasker



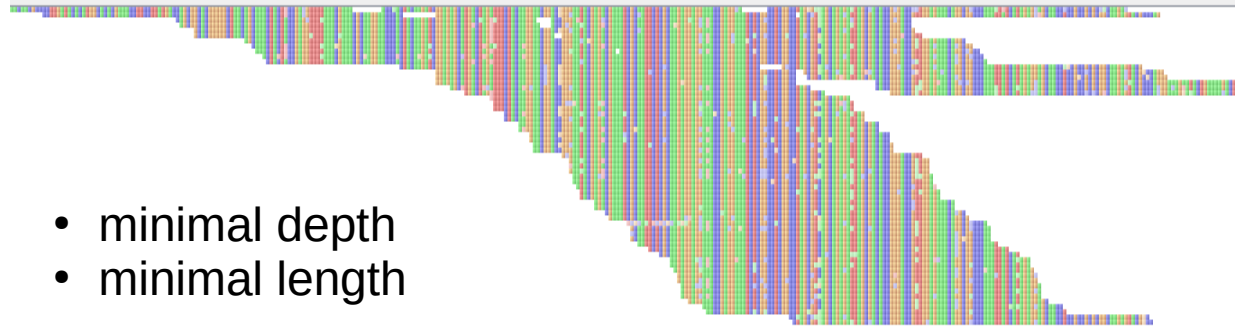
- Repbase
- DFAM
- **Custom library**

Library Based Repeat Annotation



- Custom library

Contigs from clustering results



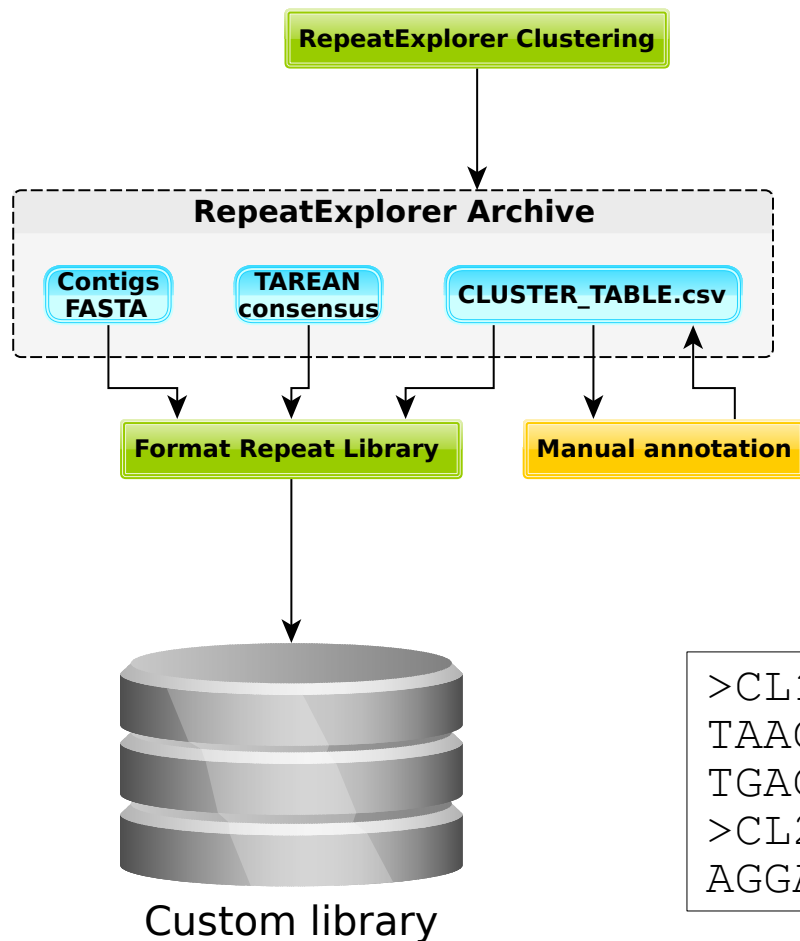
- minimal depth
- minimal length

TAREAN consensus

cAGTAGAATGATTTTCTATCTTATGACTCTTGGGAAAAATGGAACTGATTTTTCGAATATTTAGAGTCTAAAAACCTTACATTTCAGAAATCTCAGA

- as dimers

Library Based Repeat Annotation



Library preparation

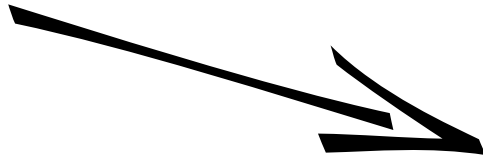
- Contig filtering
- Only clusters described in cluster table are used for library
- Fasta header - hierarchical classification

```
>CL1Contig1#Class/subclass/subclass  
TAAGTAGTGTTTCCTTGTTAGAAGATACAAAGCCA  
TGACTA  
>CL2Contig4#Class/subclass/subclass  
AGGATAAGCTTGCGGTTTAAGTTCTTATACTCAAT
```

Library Based Repeat Annotation



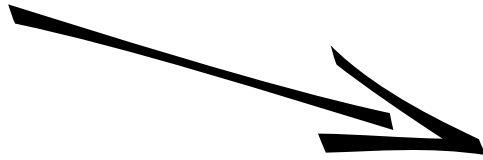
RepeatMasker



Library Based Repeat Annotation



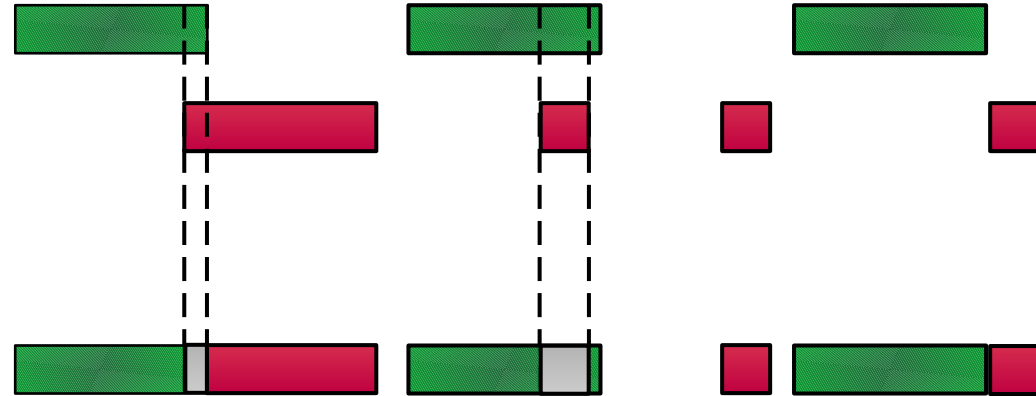
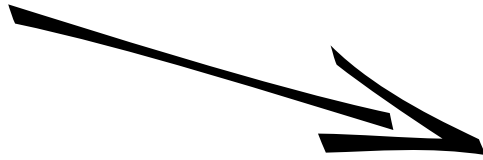
RepeatMasker



Library Based Repeat Annotation



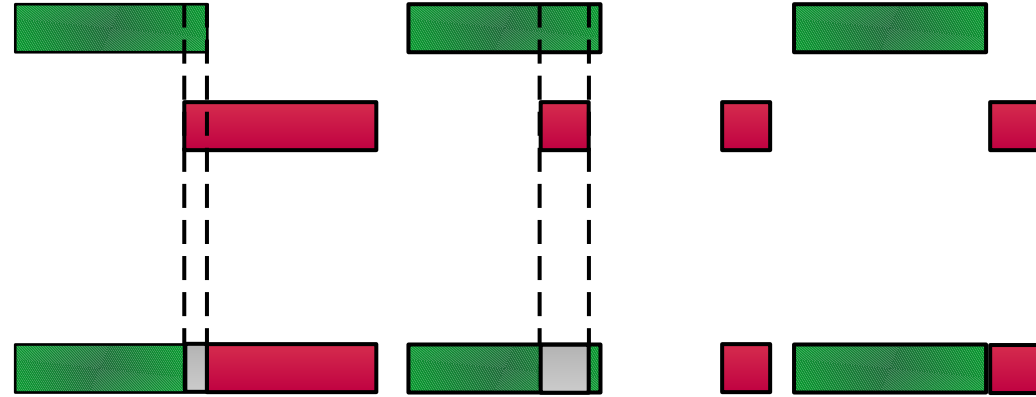
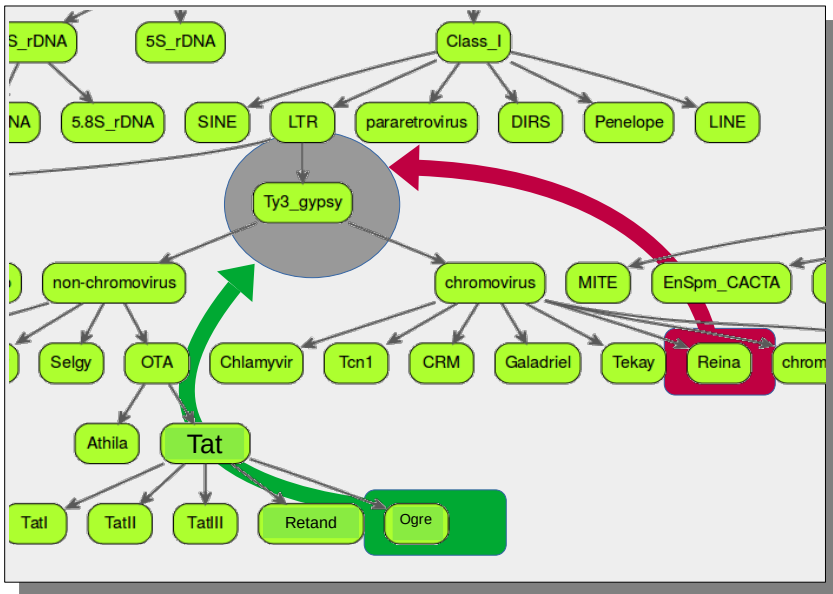
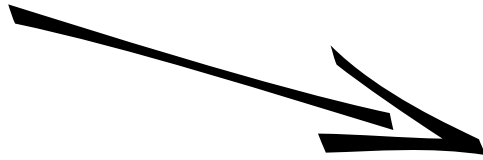
RepeatMasker



Library Based Repeat Annotation



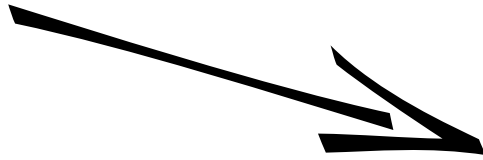
RepeatMasker



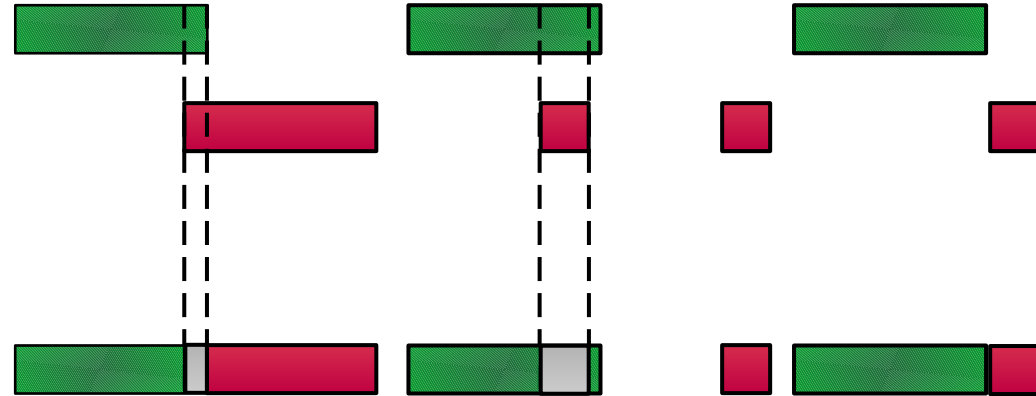
Library Based Repeat Annotation



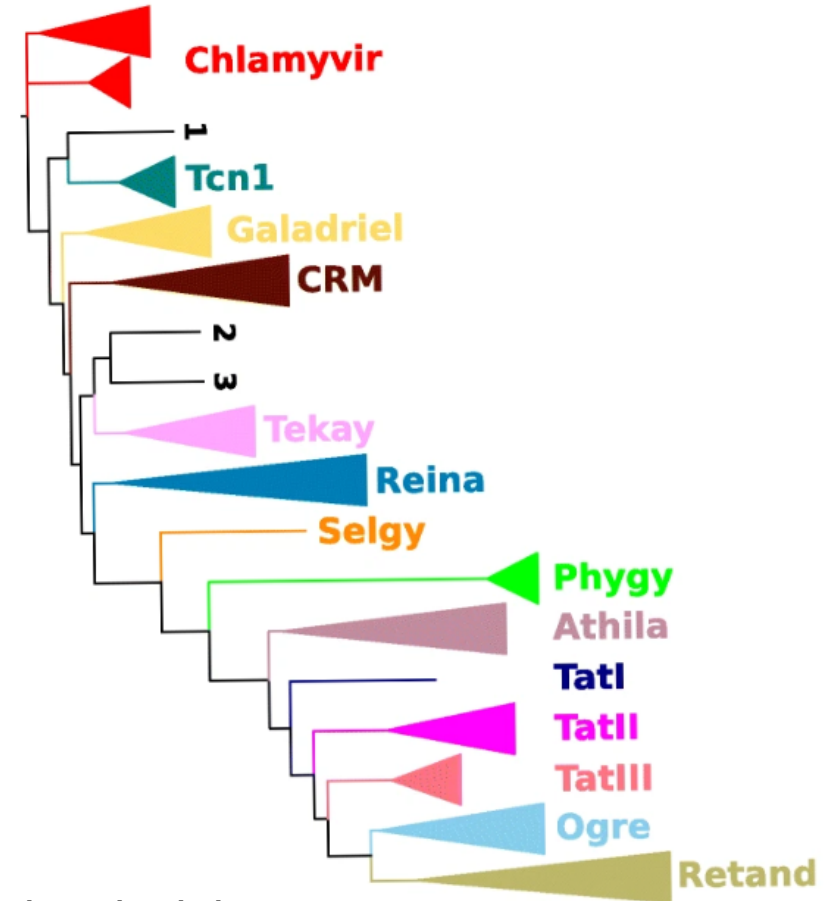
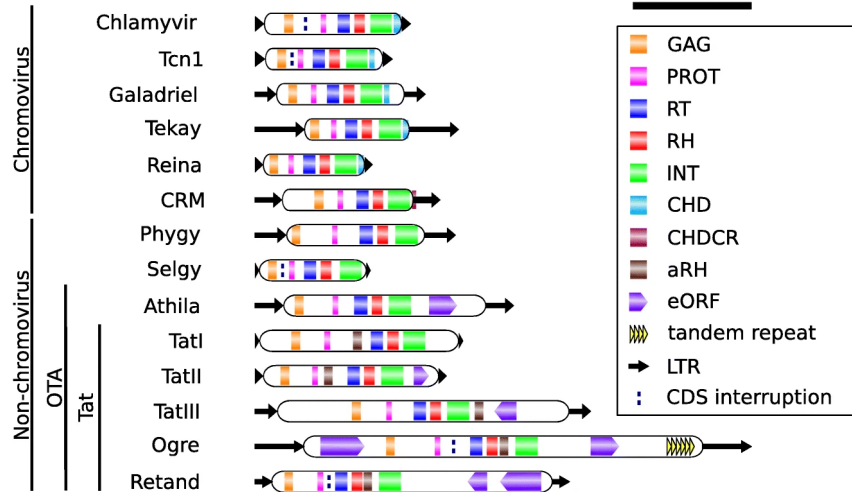
RepeatMasker



- GFF3 output
- Hierarchical user provided classification
- Information about conflicts is kept

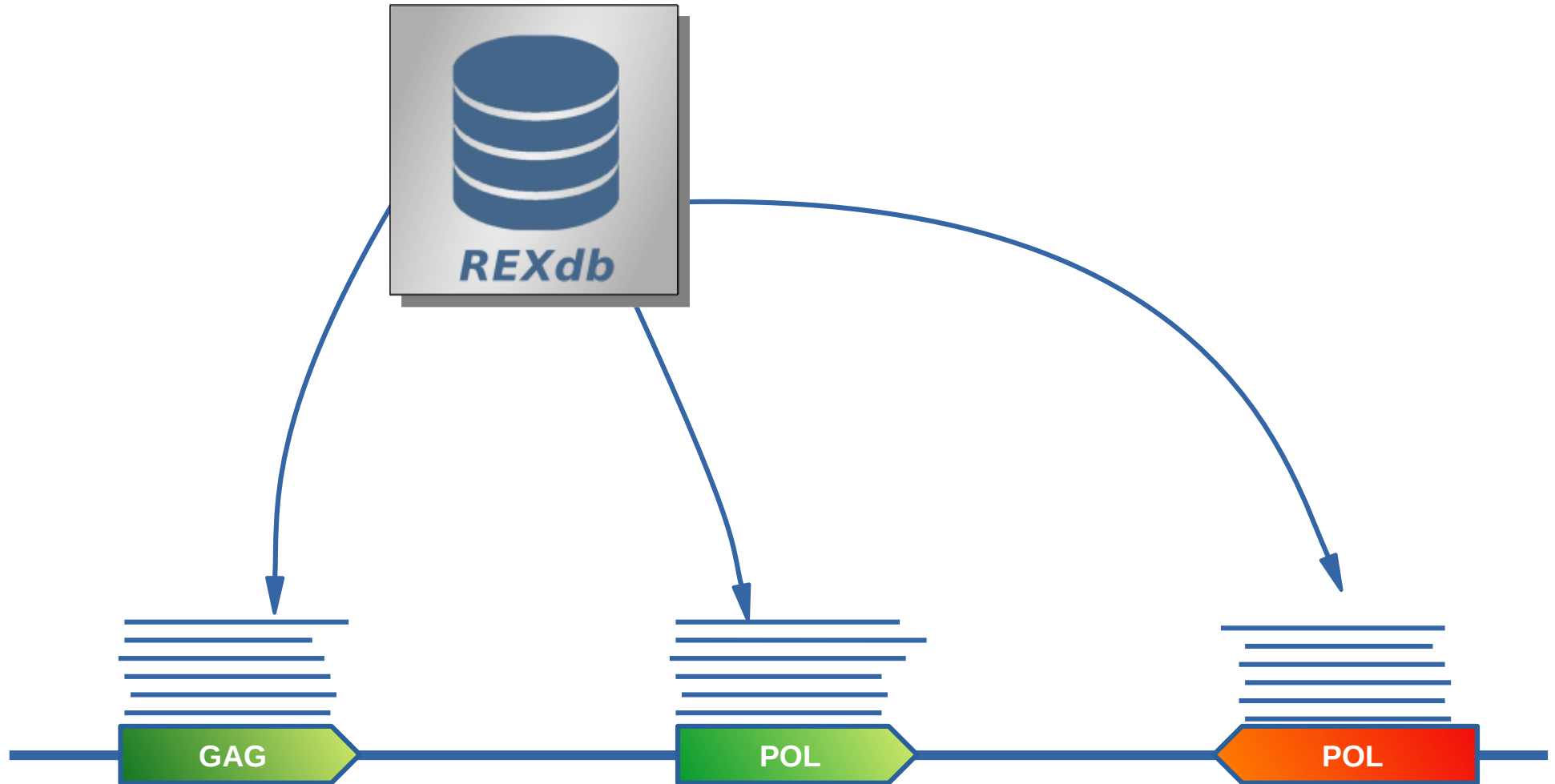


DANTE – domain based annotation of transposable elements

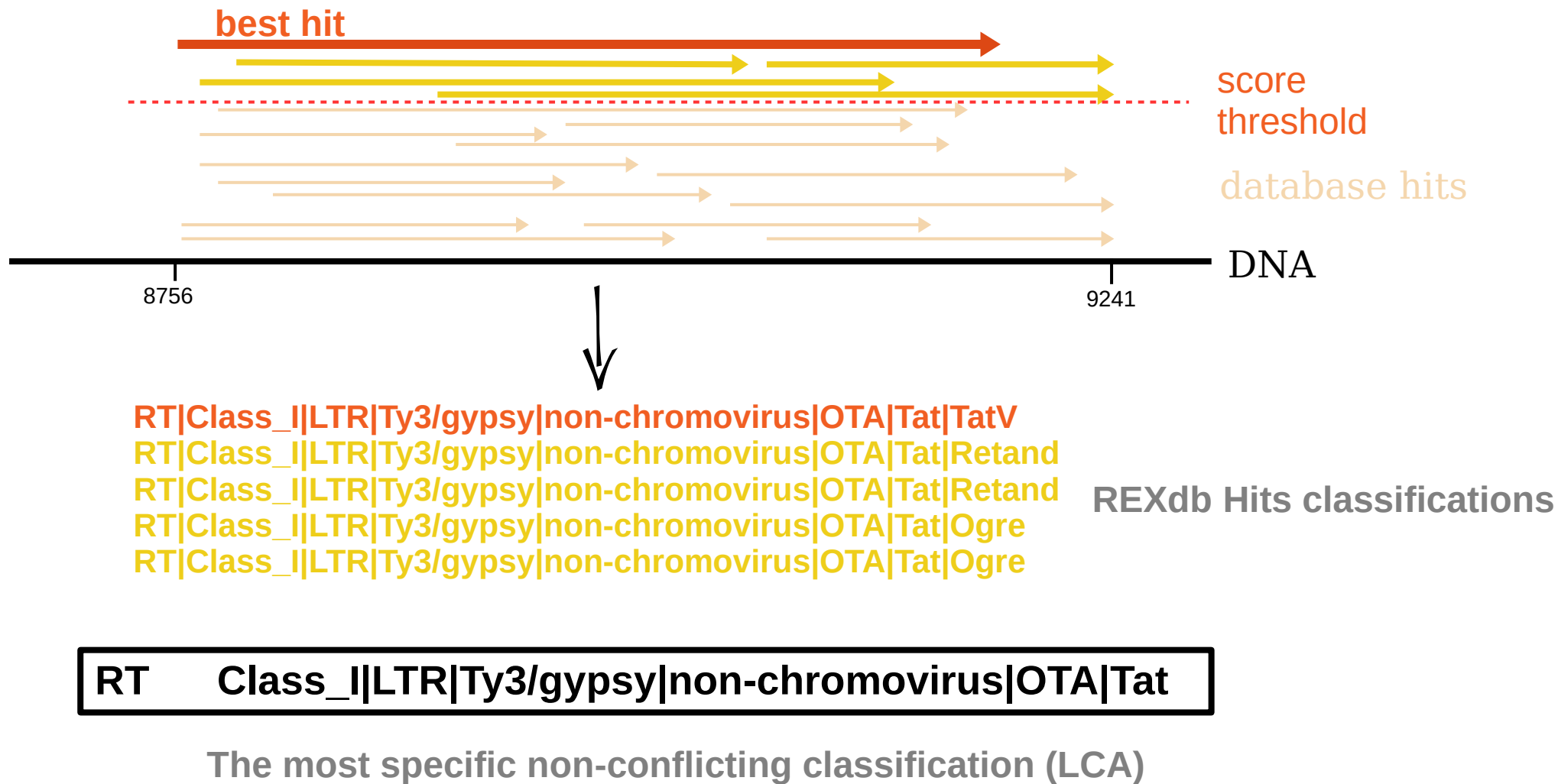


Annotation and classification based on phylogenetic principle

DANTE – domain based annotation of transposable elements



DANTE – domain based annotation of transposable elements



DANTE – domain based annotation of transposable elements



DANTE – domain based annotation of transposable elements



Custom
library
annotation



DANTE
annotation

DANTE – domain based annotation of transposable elements



Custom
library
annotation



DANTE
annotation

DANTE LTR – structure based annotation

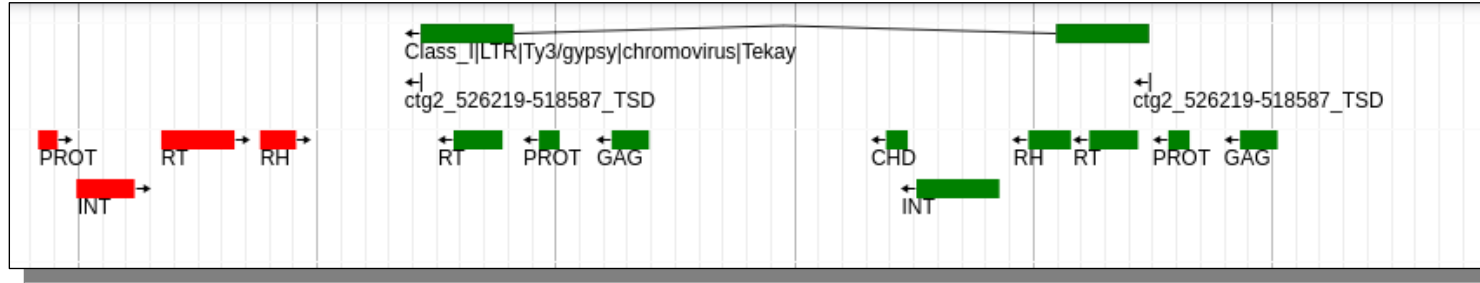


Suitable tools?

- LTR_Finder
 - LTRharvest
 - LTRdigest
 - LTRDetector
 - LTR_STRUCT
- extremely slow
 - high number of false positive
 - false negative
 - limited protein domain database (Pfam, PROSITE, Uniprot)

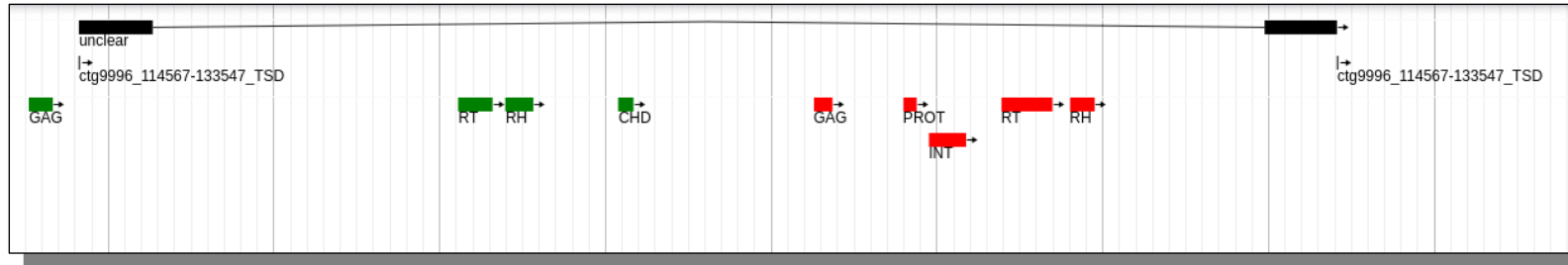


DANTE LTR – structure based annotation



LTR_Finder

DANTE



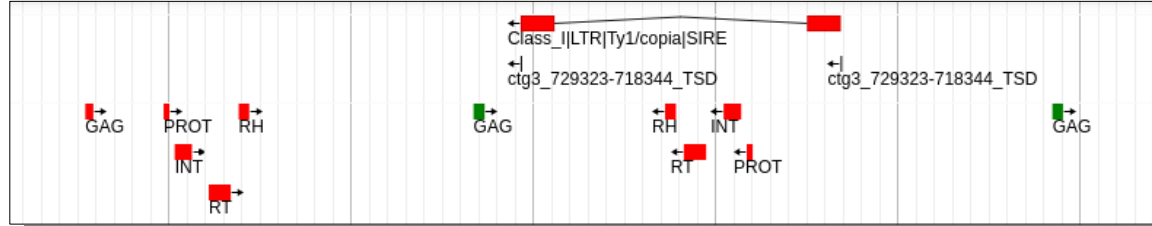
LTR_Finder

DANTE

Ty3/gypsy

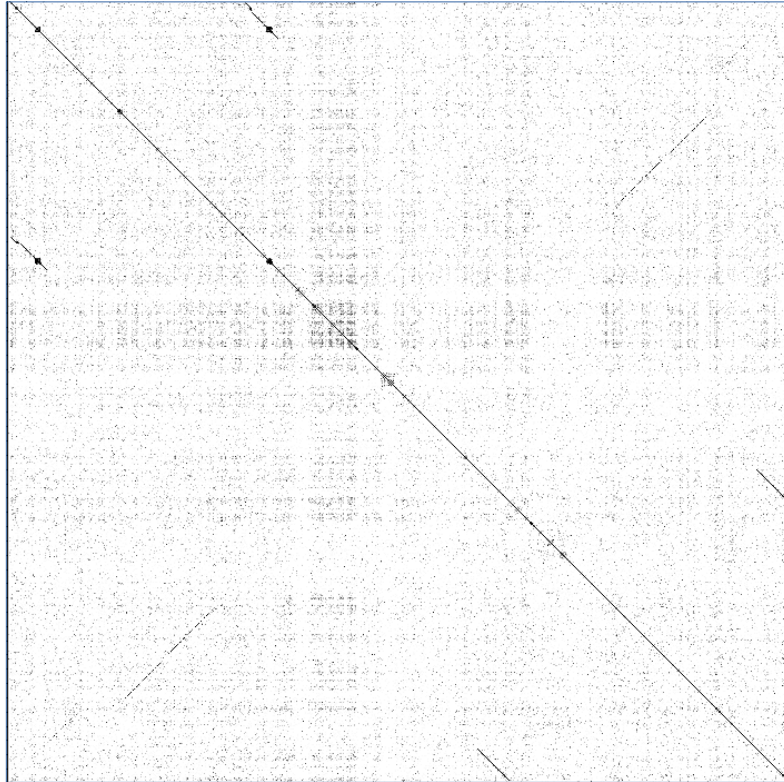
Ty1/copia

DANTE LTR – structure based annotation

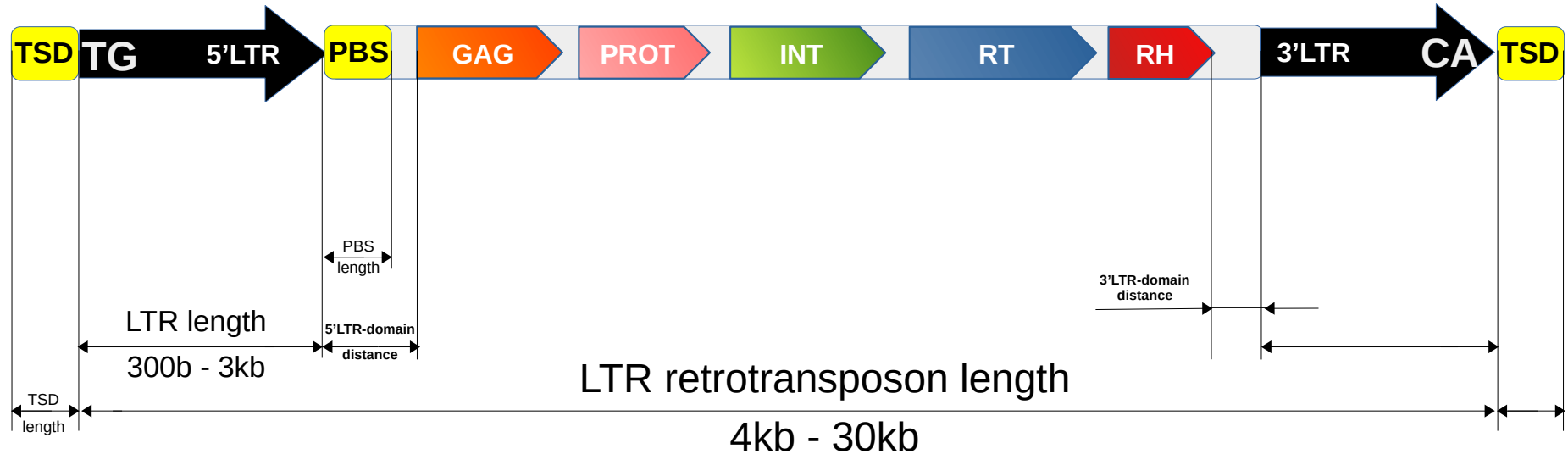


LTR_Finder

DANTE



DANTE LTR – structure based annotation



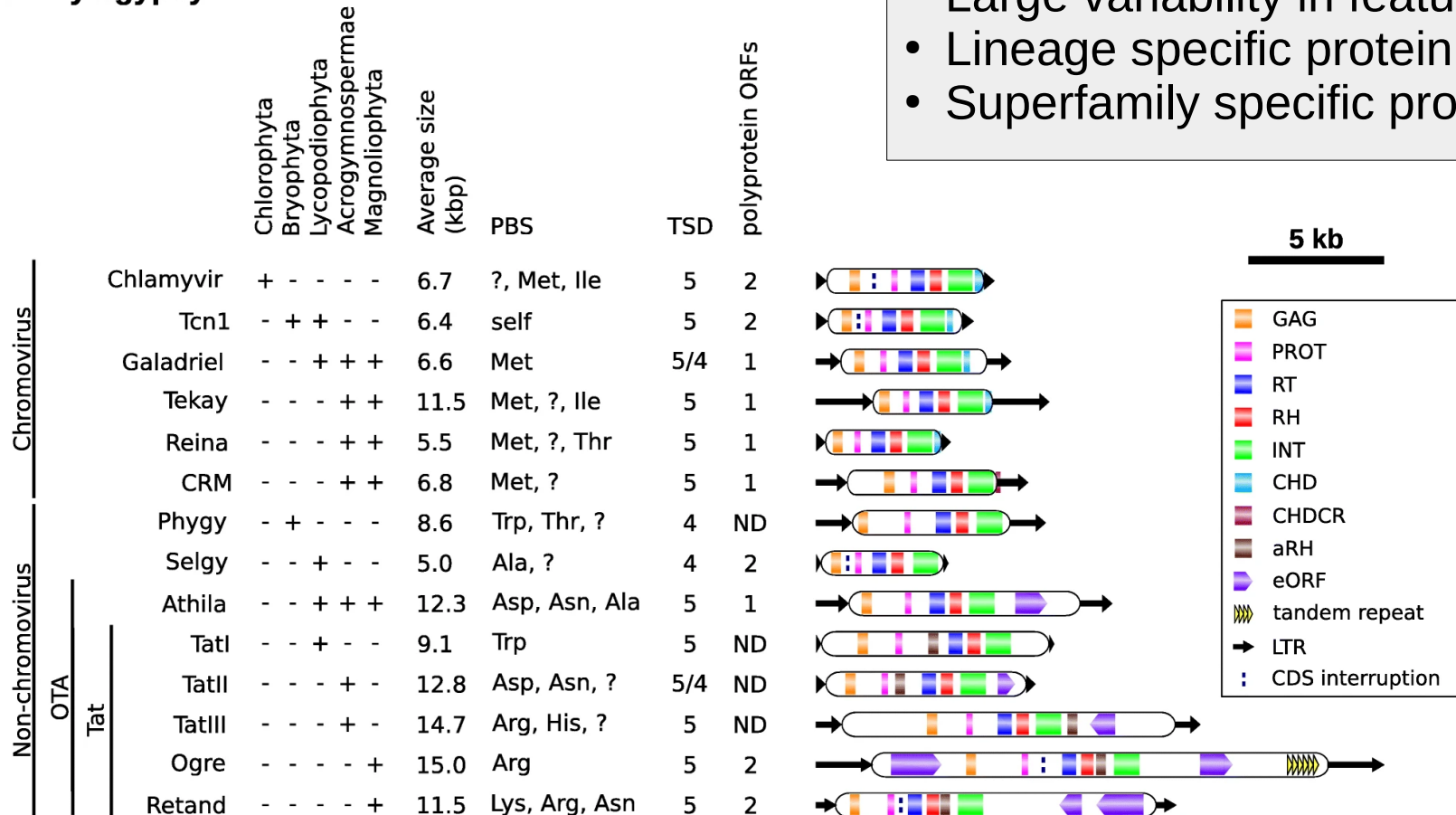
Search constraints:

- Feature length (min, max)
- Presence of features (PBS, PPT, TSD, TG/CA, domains)

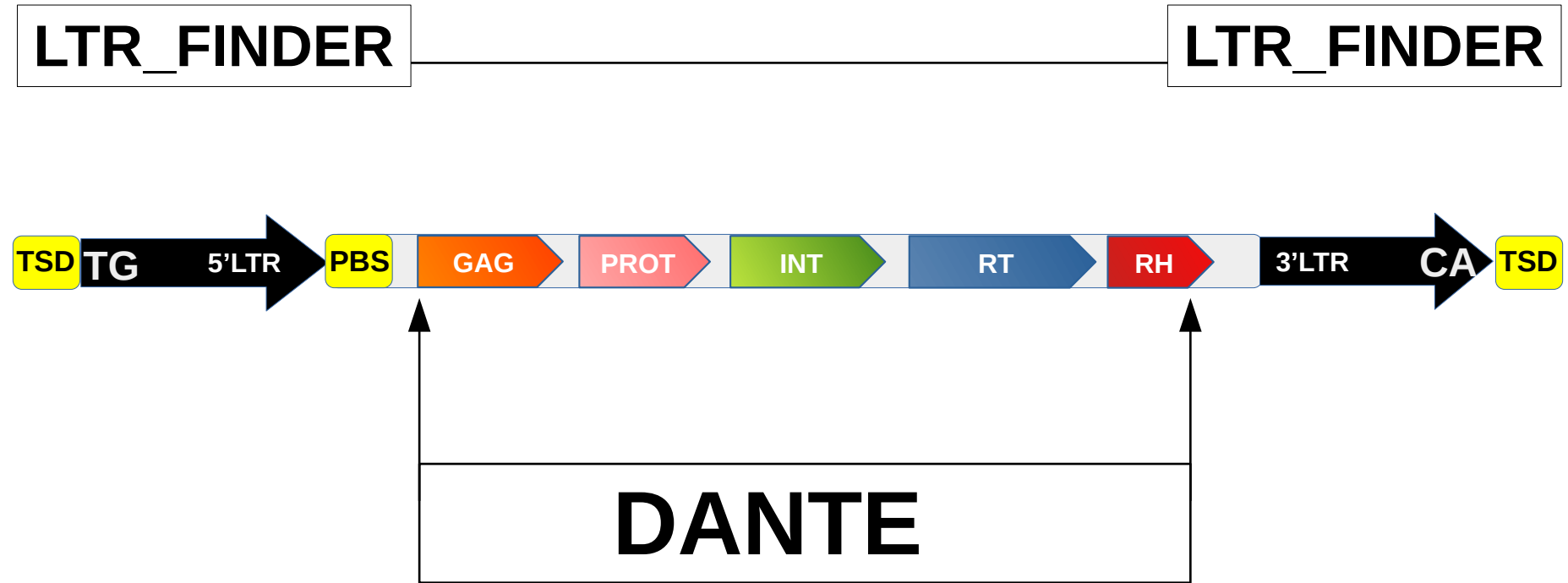
DANTE LTR – structure based annotation

A Ty3/gypsy

- Large variability in feature lengths
- Lineage specific protein domains
- Superfamily specific protein domains order



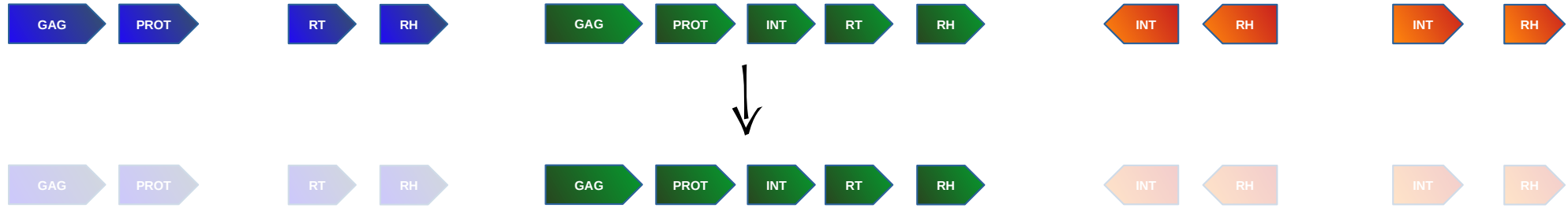
DANTE LTR – structure based annotation



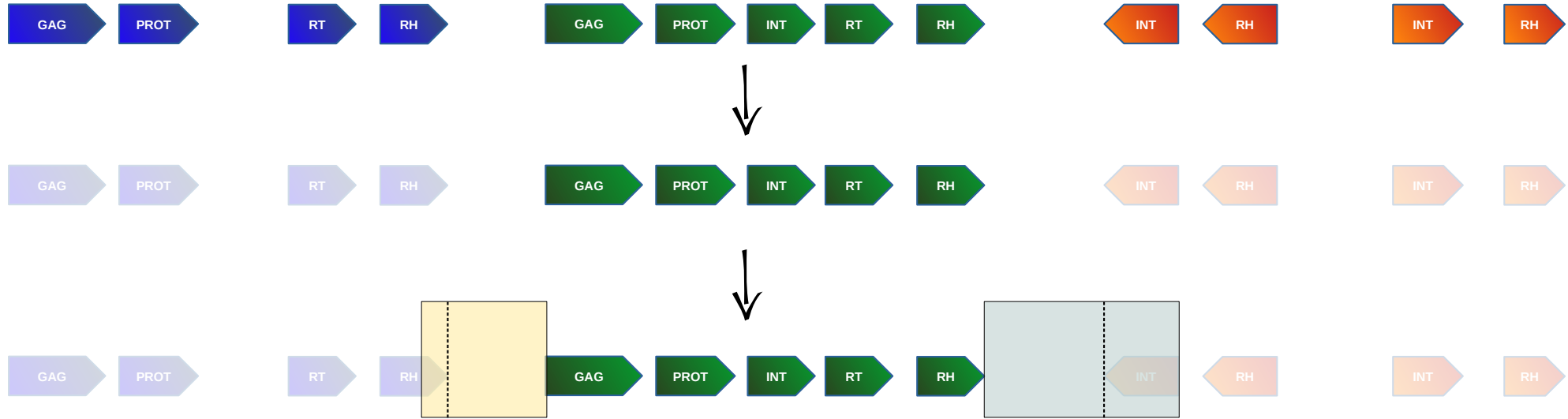
DANTE LTR – WORKFLOW



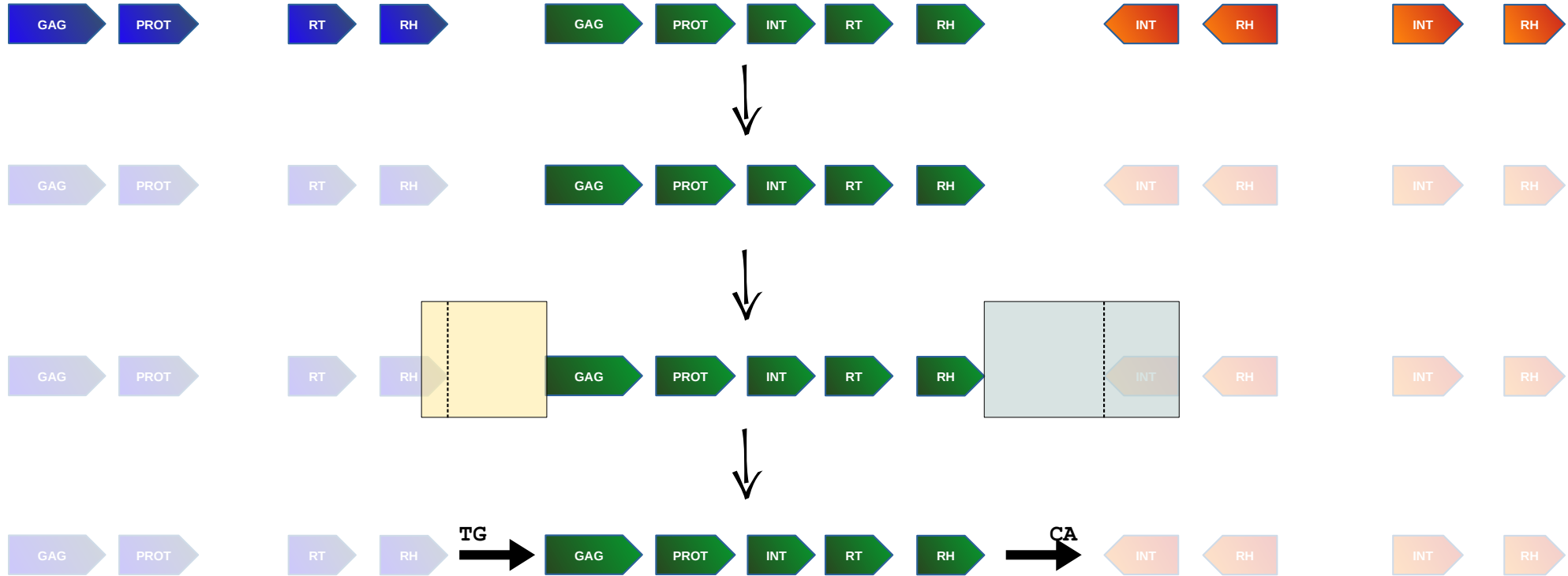
DANTE LTR – WORKFLOW



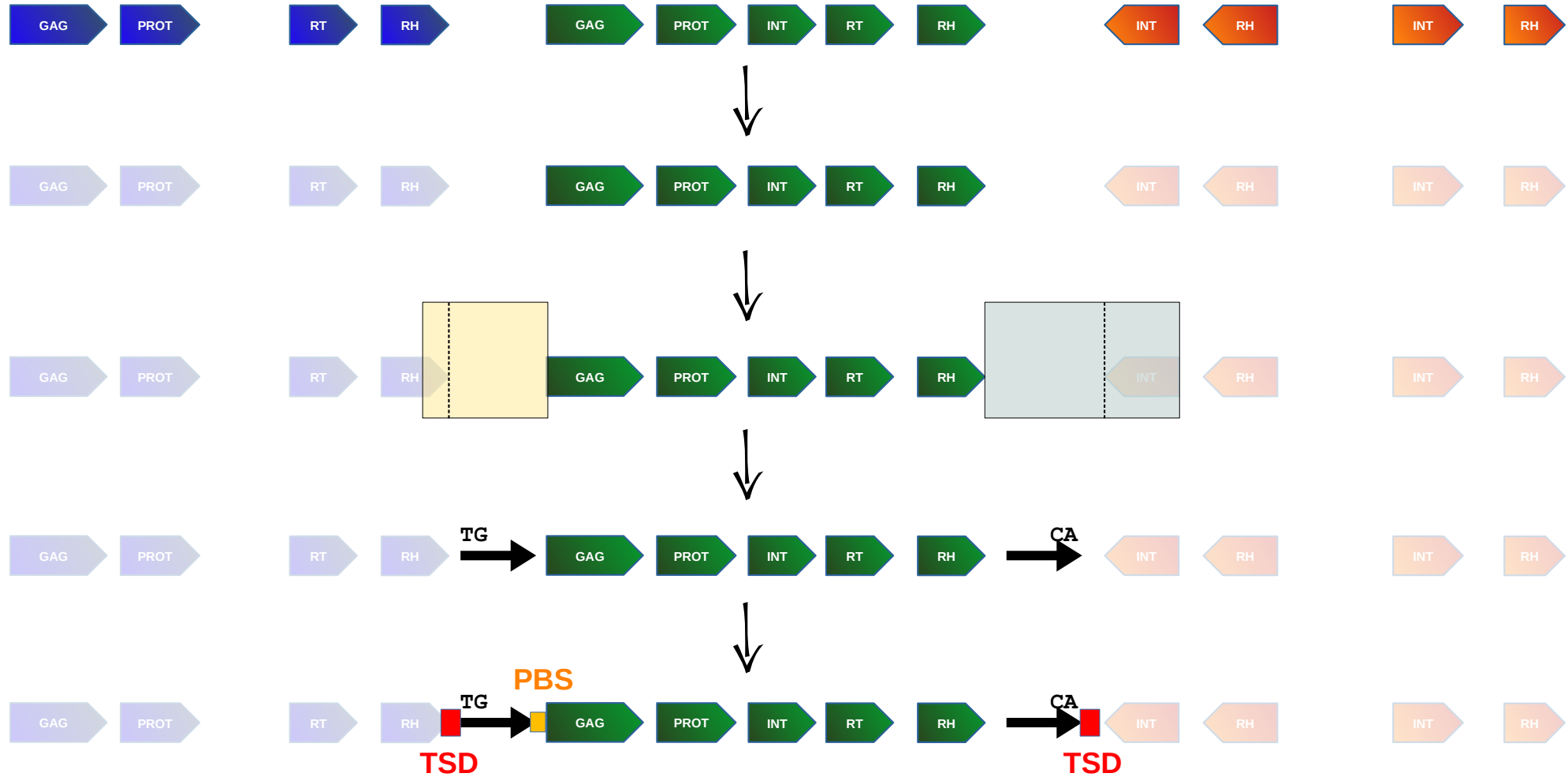
DANTE LTR – WORKFLOW



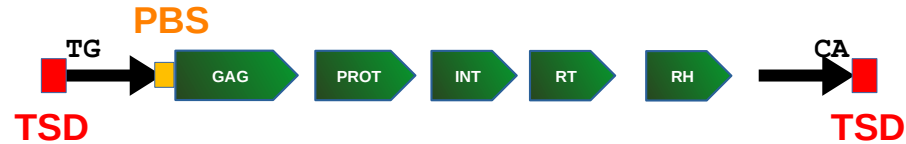
DANTE LTR – WORKFLOW



DANTE LTR – WORKFLOW



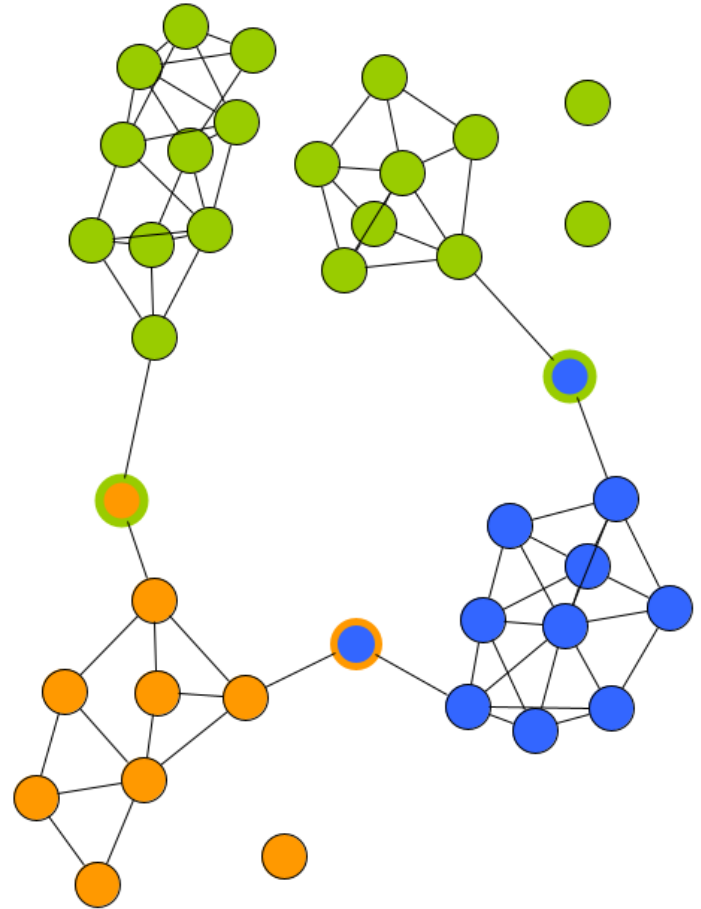
DANTE LTR – Retrotransposons ranks



Rank	Annotation
DLTP	Elements with identified protein Domains, 5'LTR, 3'LTR, TSD and PBS
DLP	Elements with identified protein Domains, 5'LTR, 3'LTR and PBS (TSD was not found)
DLT	Elements with identified protein Domains, 5'LTR, 3'LTR and TSD (PBS was not found)
DL	Elements with protein Domains, 5'LTR and 3'LTR (PBS and LDS were not found)

DANTE LTR – Filtering

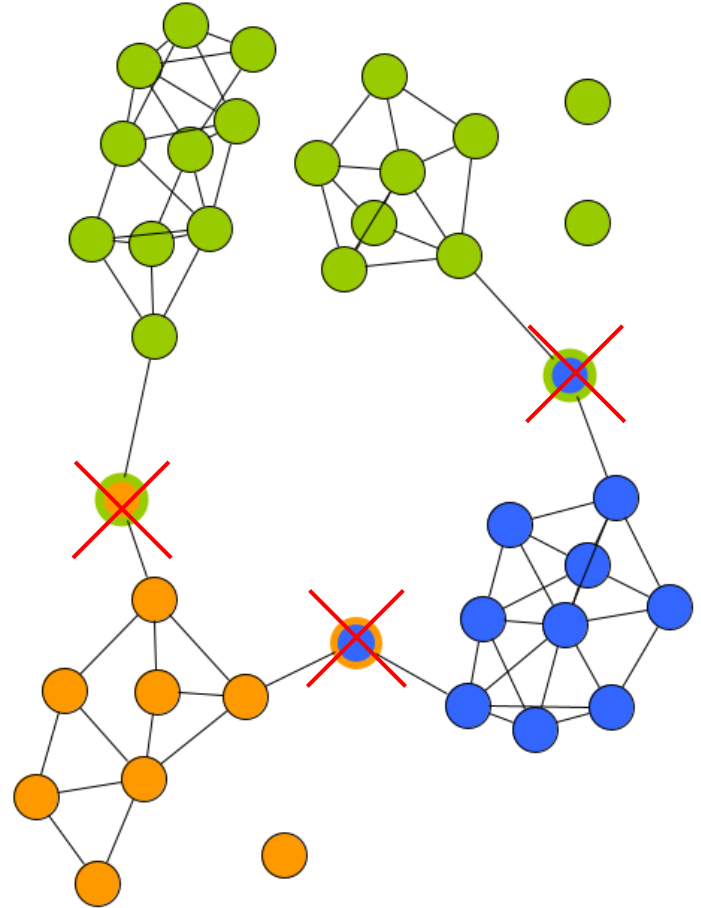
All-to-All sequence comparison



DANTE LTR – Filtering

All-to-All sequence comparison

Cross-lineage similarities

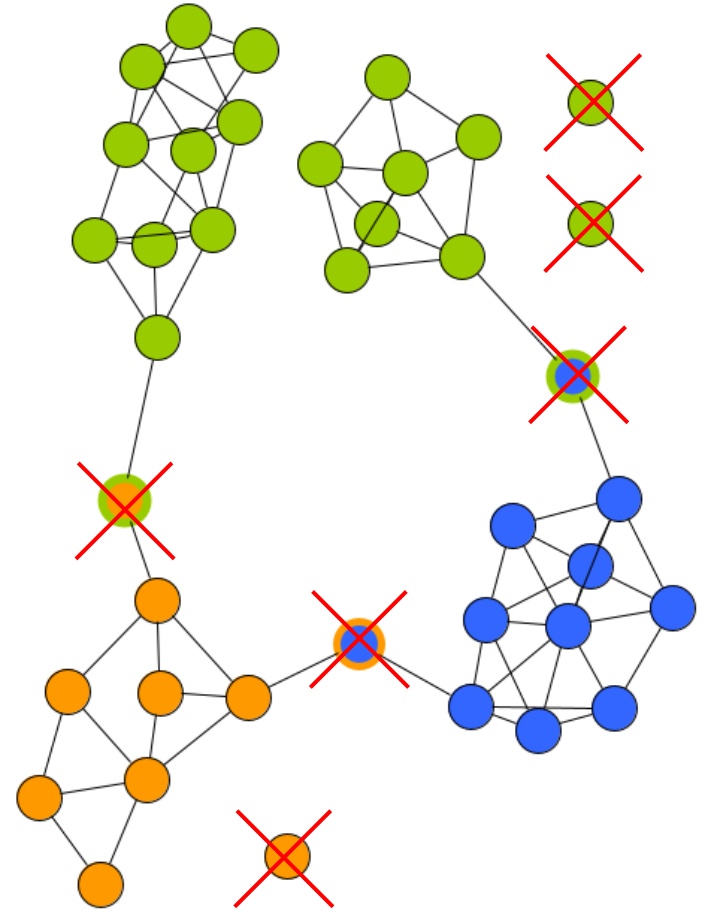


DANTE LTR – Filtering

All-to-All sequence comparison

Cross-lineage similarities

Minimal copy number (DLP, DLT)



DANTE LTR - Output

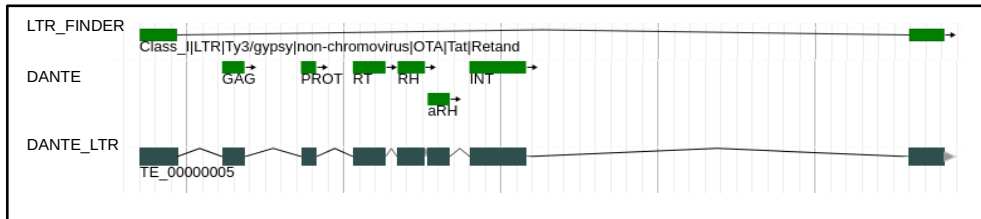
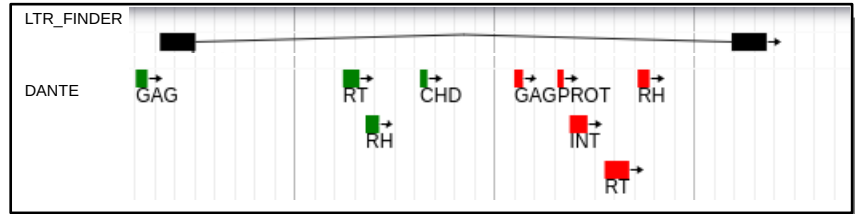
	SOURCE	TYPE	START	END			ATTRIBUTES
1	dante_ltr	transposable_element	3780765	3785720	.	+	ID=TE_00000001;LTR_Identity=100;LTR5_length=440;LTR3_length=440;TSD=CTTGT;Final_Classification=Class_I LTR Ty1/copia Ivana;Region_Hits_Classifications=NA;trna_id=ATCAAAACCTAGCTCTGATAcca__Met-3x;Rank=DLTP
1	dante_ltr	long_terminal_repeat	3785281	3785720	.	+	LTR_Identity=100;Final_Classification=Class_I LTR Ty1/copia Ivana;LTR=3LTR;Parent=TE_00000001;Region_Hits_Classifications=NA;Rank=DLTP
1	dante_ltr	long_terminal_repeat	3780765	3781204	.	+	LTR_Identity=100;Final_Classification=Class_I LTR Ty1/copia Ivana;LTR=5LTR;Parent=TE_00000001;Region_Hits_Classifications=NA;Rank=DLTP
1	dante	protein_domain	3781451	3781729	498	+	Final_Classification=Class_I LTR Ty1/copia Ivana;Parent=TE_00000001;Name=GAG;Region_Hits_Clas
1	dante	protein_domain	3782237	3782452	406	+	Final_Classification=Class_I LTR Ty1/copia Ivana;Parent=TE_00000001;Name=PROT;Region_Hits_Cla
1	dante	protein_domain	3782639	3783238	1132	+	Final_Classification=Class_I LTR Ty1/copia Ivana;Parent=TE_00000001;Name=INT;Region_Hits_Clas
1	dante	protein_domain	3783782	3784549	1448	+	Final_Classification=Class_I LTR Ty1/copia Ivana;Parent=TE_00000001;Name=RT;Region_Hits_Class
1	dante	protein_domain	3784817	3785197	728	+	Final_Classification=Class_I LTR Ty1/copia Ivana;Parent=TE_00000001;Name=RH;Region_Hits_Class
1	dante_ltr	target_site_duplication	3785721	3785725	.	.	Parent=TE_00000001;Region_Hits_Classifications=NA;Rank=DLTP
1	dante_ltr	target_site_duplication	3780760	3780764	.	.	Parent=TE_00000001;Region_Hits_Classifications=NA;Rank=DLTP
1	dante_ltr	primer_binding_site	3781208	3781220	.	+	Parent=TE_00000001;Region_Hits_Classifications=NA;trna_id=ATCAAAACCTAGCTCTGATAcca__Met-3x;Ran

DANTE LTR vs. LTR_FINDER

	LTR_FINDER	DANTE_LTR (unfiltered)	DANTE_LTR (clean)
Total	10975	8982	4515
With protein domain(s)	Single lineage	8982	4515
	Multiple lineages (conflict)	-	-

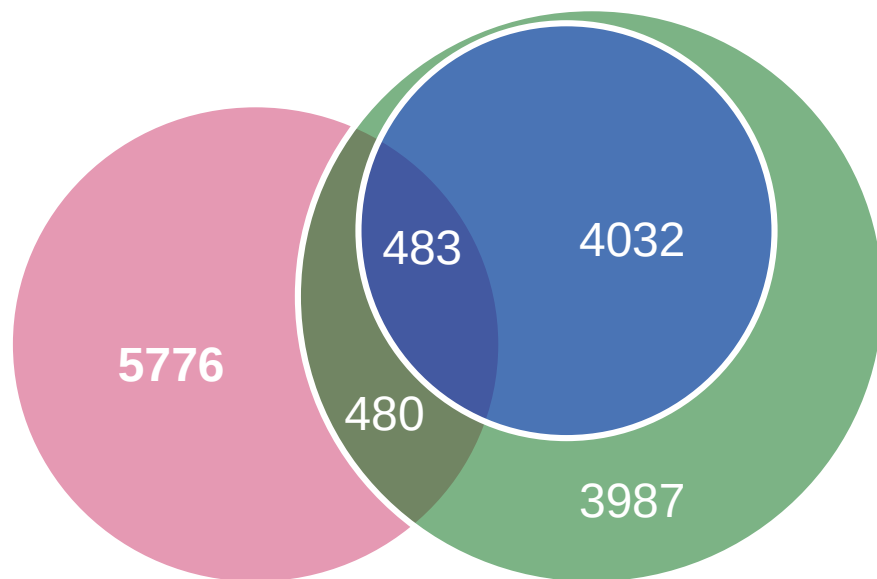
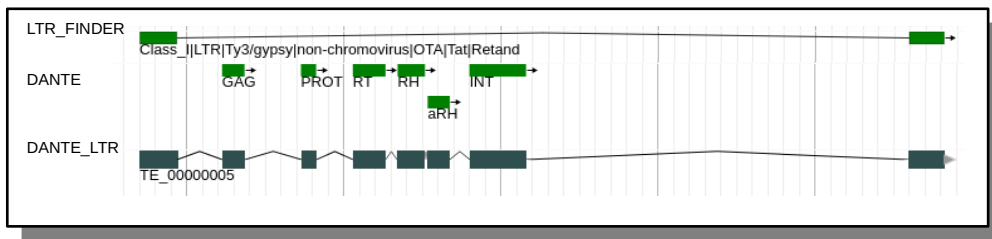
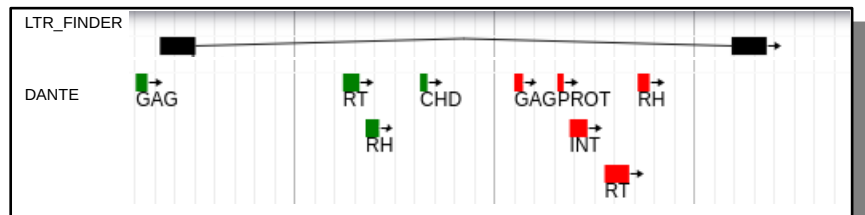
DANTE LTR vs. LTR_FINDER

	LTR_FINDER	DANTE_LTR (unfiltered)	DANTE_LTR (clean)
Total	10975	8982	4515
Single lineage	6739	8982	4515
With protein domain(s) Multiple lineages (conflict)	281	-	-

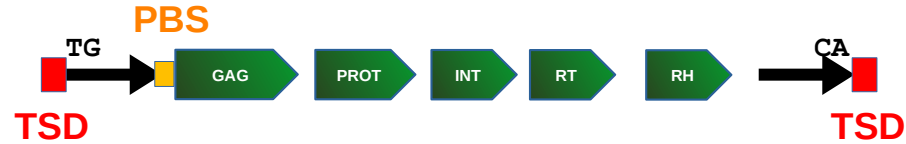


DANTE LTR vs. LTR_FINDER

	LTR_FINDER	DANTE_LTR (unfiltered)	DANTE_LTR (clean)
Total	10975	8982	4515
Single lineage	6739	8982	4515
With protein domain(s)			
Multiple lineages (conflict)	281	-	-

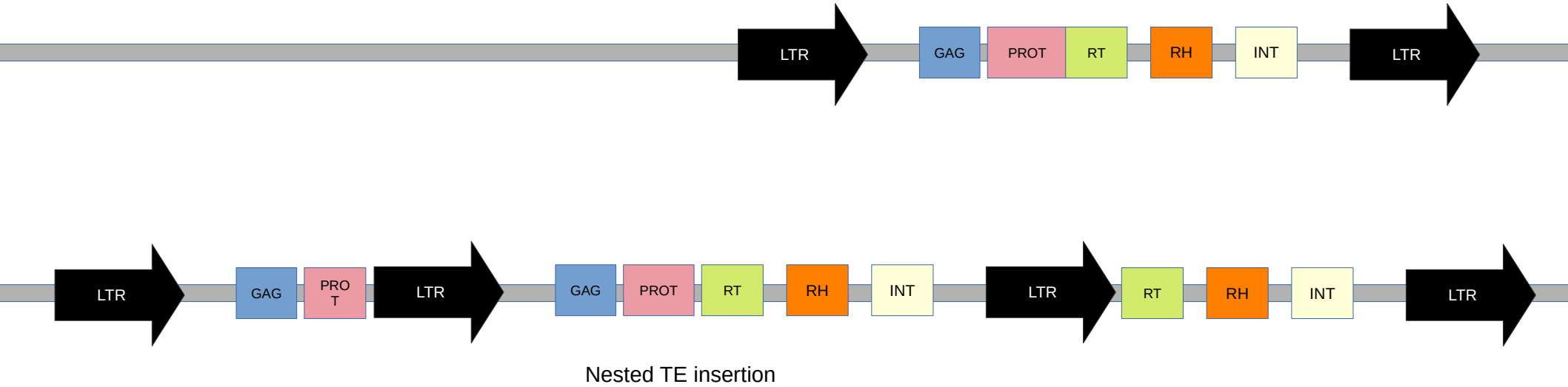


DANTE LTR – Applications

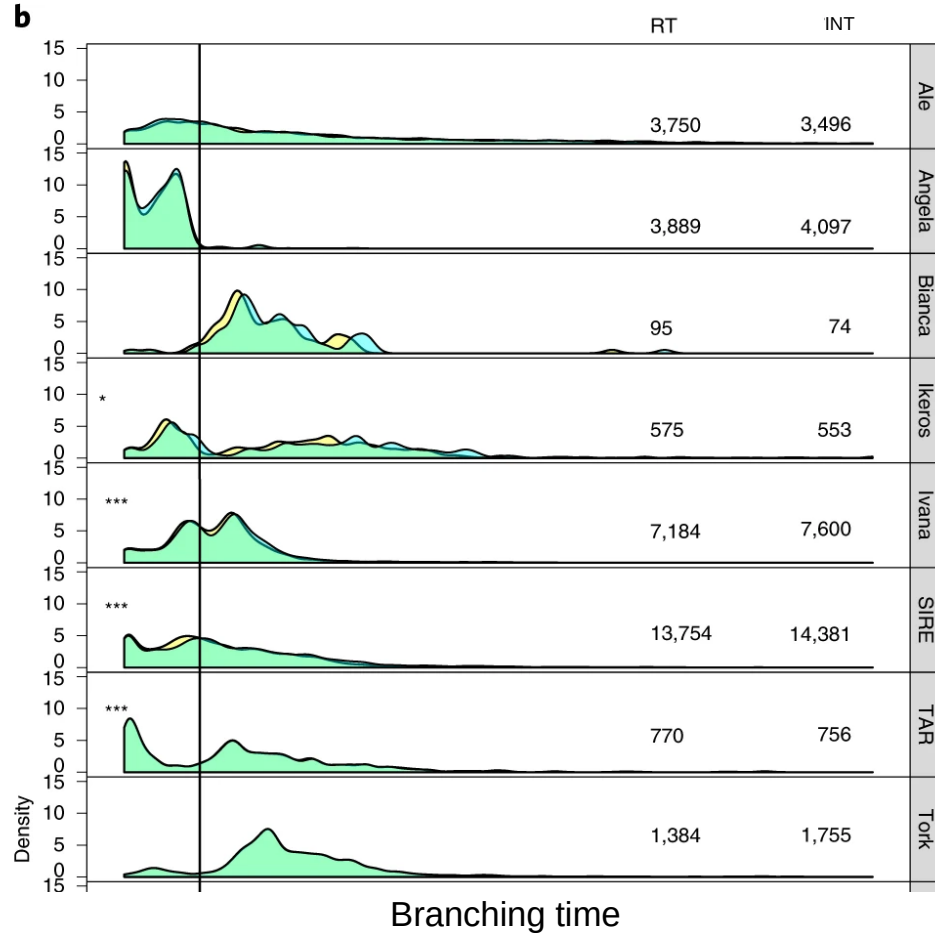


- Custom library for assembly annotation
- Dating of retrotransposons activity ???

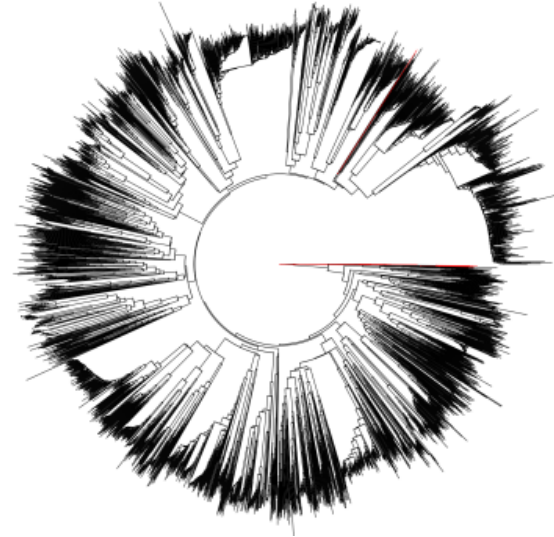
Dating of retrotransposons activity



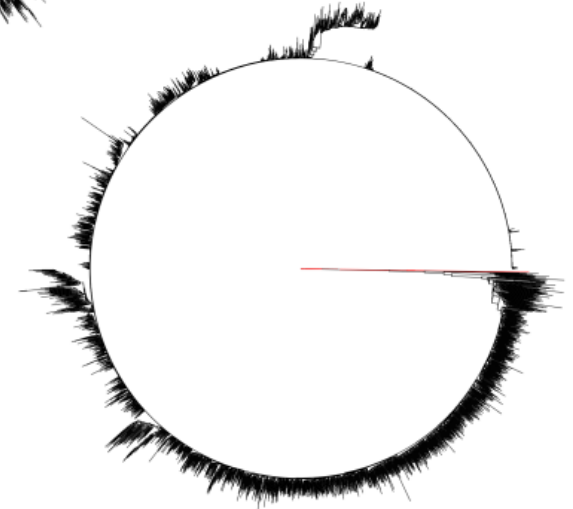
Dating of retrotransposons activity



Ty1/Copia Ale

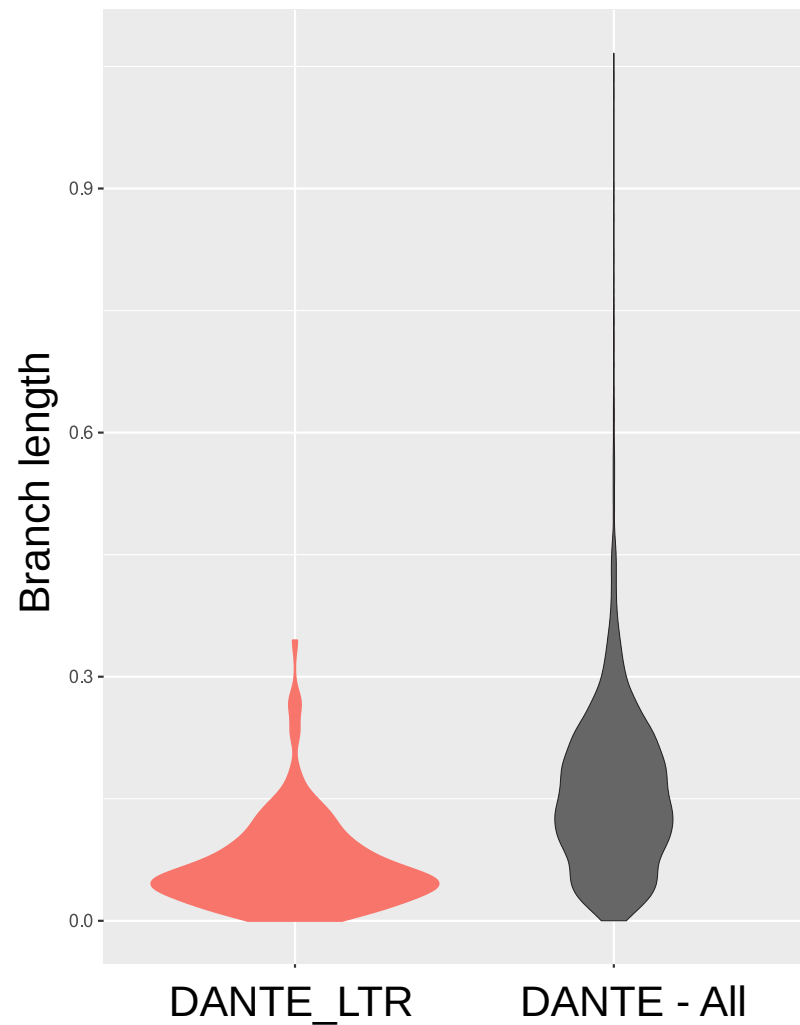
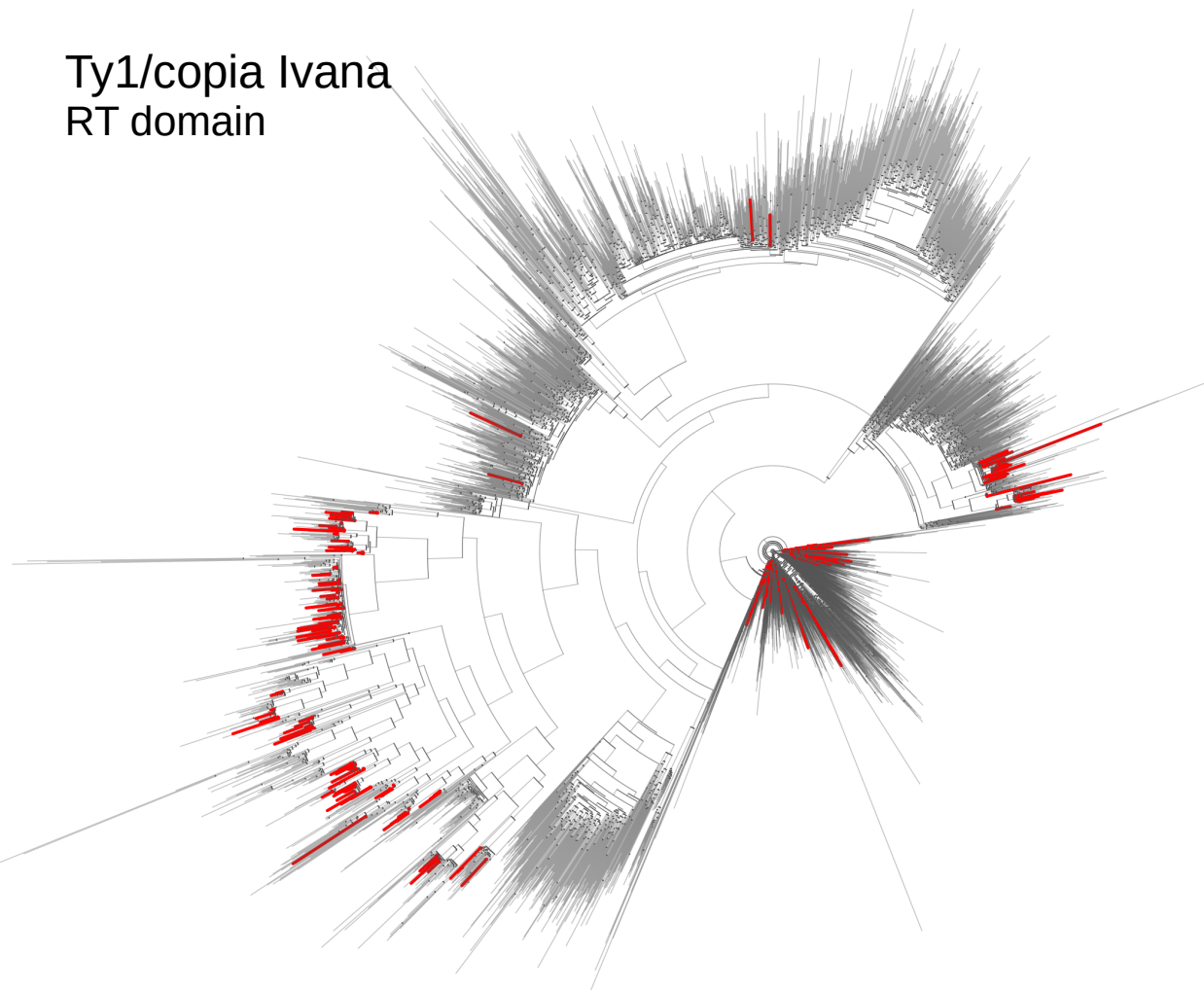


1/Copia Angela



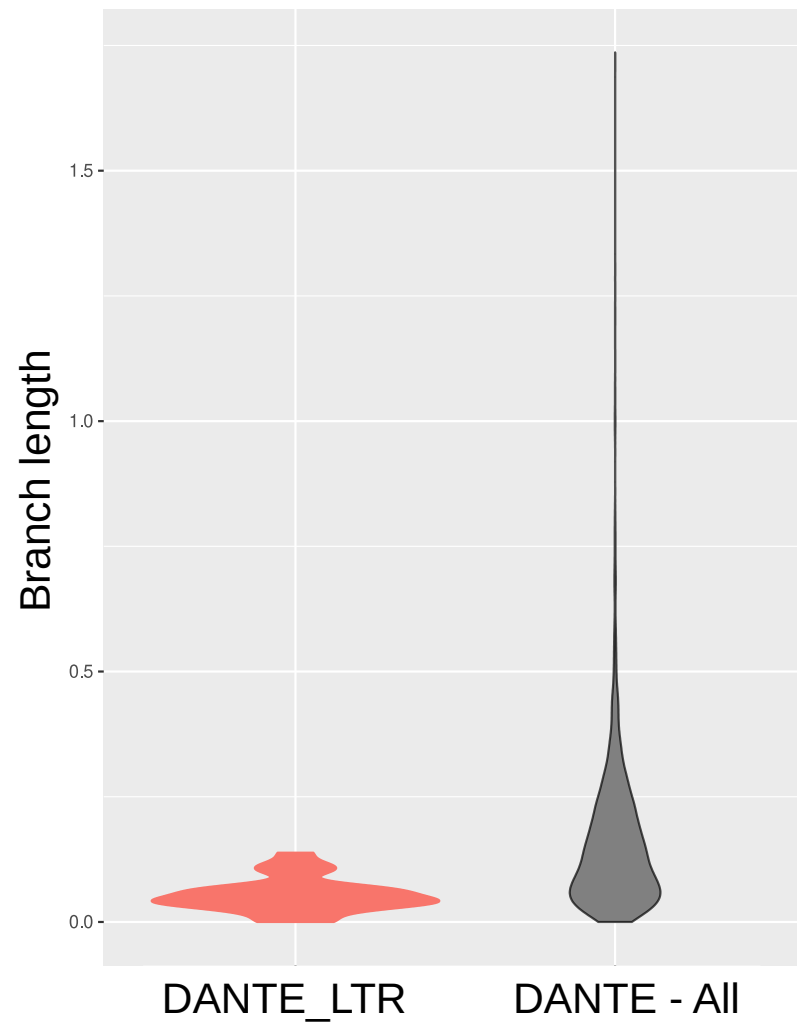
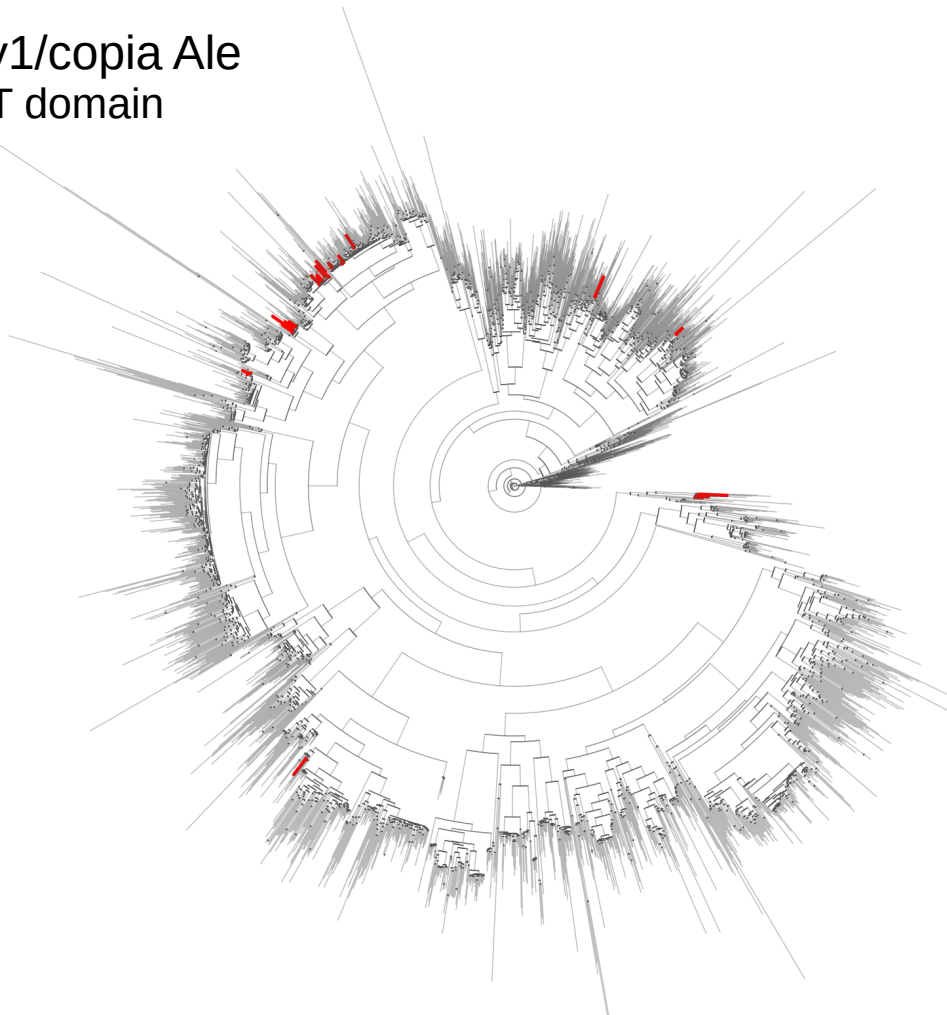
DANTE LTR

Ty1/copia Ivana
RT domain



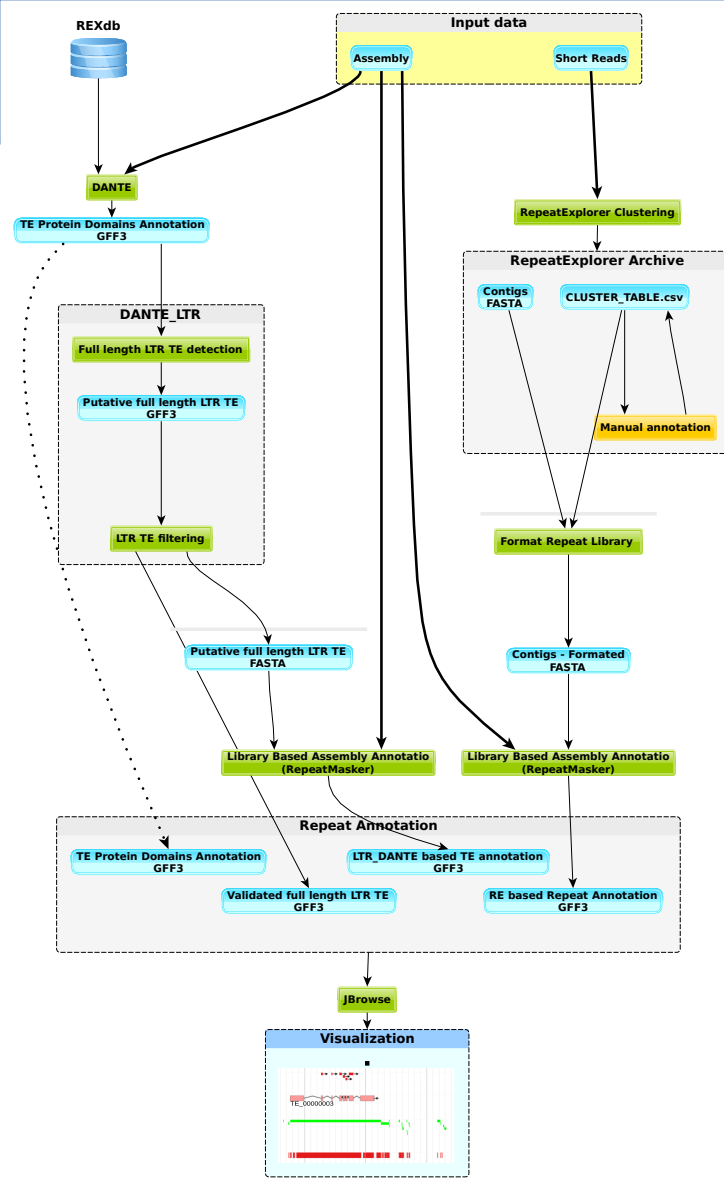
DANTE LTR

Ty1/copia Ale
RT domain



Complete workflow

- RepeatExplorer to create custom library
- Library based annotation using RepeatMasker
- DANTE
- DANTE_LTR to identify intact LTR retrotransposons
- Use intact elements as second custom library



Acknowledgment



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Petr Novak



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Ivana Krenkova
Martin Demko

